THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — SCIENCE · HARD LEVEL CHAPTER TEST SET 2

Chapter 11: **Patterns in Life: Diversity and Classification** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. Assertion (A): Fungi are placed in a separate kingdom. Reason (R): They are autotrophic like plants. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

2. An animal has a segmented body, a coelom and a closed circulatory system without jointed legs. It is a(n): 1

(a) annelid **(b)** nematode **(c)** mollusc **(d)** arthropod

3. The statement that the lion and the tiger belong to the same genus means that they share: 1

(a) the species name **(b)** the genus name Panthera
(c) the same habitat only **(d)** the common name

4. A cell with no nucleus but with a cell wall belongs to: 1

(a) Fungi **(b)** Plantae **(c)** Protista **(d)** Monera

5. A student classifies the penguin as a fish because it swims. The best correction is: 1

(a) it is a reptile
(b) it is a bird as it has feathers and lays eggs
(c) it is a mammal
(d) it is a fish

6. Assertion (A): Viruses are not placed in any of the five kingdoms. Reason (R): Viruses are non-cellular. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

7. An animal has a bilaterally symmetrical, unsegmented body, a muscular foot and a mantle. It is a: 1

(a) nematode **(b)** mollusc **(c)** annelid **(d)** echinoderm

8. Which of the following correctly lists the rank 'family' for humans? 1

(a) Felidae **(b)** Bovidae **(c)** Hominidae **(d)** Canidae

9. **Assertion (A):** Birds and mammals are warm-blooded animals. 1
Reason (R): All vertebrates are warm-blooded.
Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

10. **Assertion (A):** Fungi are autotrophs. 1
Reason (R): The cell wall of fungi is made of chitin.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. Why are lichens an example of symbiosis? State the partners and the benefit. 2

12. State the importance of the taxonomic key in identification of organisms. 2

13. Why is the classification of viruses debated? State two reasons. 2

14. A pea plant and a rice plant are placed in the same class Angiospermae but different subclasses. State one reason. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Compare the Porifera and Cnidaria in terms of body organisation, symmetry and cell types. 3

16. List the economically important insects with their roles. 3

17. Explain how scientists might use DNA sequences to classify organisms and improve the classification. 3

OR

The tiger and the leopard both belong to the genus Panthera while the dog belongs to the genus Canis. Explain what this shows about relationships. 3

18. State the role of decomposers (fungi and bacteria) in the ecosystem and what happens if they are removed. 3

OR

Name the common features of all chordates and how the human body shows them at different stages. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study.** A student notes the features of three animals. Animal P has a backbone, a four-chambered heart, hair on the body and mammary glands. Animal Q has feathers, forelimbs modified into wings and lays eggs with hard shells. Animal R has moist skin without scales, lays eggs in water and its young stage breathes through gills. 5
(i) Identify the class of P.
(ii) Identify the class of Q.
(iii) Identify the class of R.
(iv) A bat flies like a bird but is classed with P. Give two reasons.

20. Explain the evolution of the vertebrate heart from fish to mammals with the number of chambers and the circulation in each. 5

OR

Explain the characteristics of the classes Aves and Mammalia and the adaptations of birds for flight, penguins for swimming, and bats for flying, indicating convergence. 5

END OF QUESTION PAPER — **Total: 40 Marks**